

ANSWER SCHEME

NO.	SOLUTION	MARKS	REMARKS
PART A			
1a)	$\lim_{x \rightarrow 3} \frac{\sqrt{x^2 + 7} - \sqrt{2x + 10}}{x - 3}$ $\lim_{x \rightarrow 3} \left(\frac{\sqrt{x^2 + 7} - \sqrt{2x + 10}}{x - 3} \right) \left(\frac{\sqrt{x^2 + 7} + \sqrt{2x + 10}}{\sqrt{x^2 + 7} + \sqrt{2x + 10}} \right)$ $\lim_{x \rightarrow 3} \frac{x^2 + 7 - (2x + 10)}{(x - 3)(\sqrt{x^2 + 7} + \sqrt{2x + 10})}$ $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{(x - 3)(\sqrt{x^2 + 7} + \sqrt{2x + 10})}$ $\lim_{x \rightarrow 3} \frac{(x - 3)(x + 1)}{(x - 3)(\sqrt{x^2 + 7} + \sqrt{2x + 10})}$ $\lim_{x \rightarrow 3} \frac{(x + 1)}{(\sqrt{x^2 + 7} + \sqrt{2x + 10})} = \frac{3 + 1}{\sqrt{3^2 + 7} + \sqrt{2(3) + 10}} = \frac{1}{2}$	K1 K1 K1 K1 J1	Multiply with the correct conjugate Simplify numerator Factorize Sub limit and answer
1b)	$f(x) = \begin{cases} \frac{x^2 - a^2}{x - a}, & x < a \\ x + 2, & x \geq a \end{cases}$ $\lim_{x \rightarrow a^+} x + 2 = a + 2$	K1	Finding limit from right

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	$\lim_{x \rightarrow a^-} \frac{x^2 - a^2}{x - a} = \lim_{x \rightarrow a^-} \frac{(x+a)(x-a)}{x-a} = \lim_{x \rightarrow a^-} x + a = a + a = 2a$ $a + 2 = 2a \quad a = 2$	K1 K1J1	Finding limit from left Equating limit from right and left and solving for value of a
Total		9 marks	
2	$5e^{3y} = x^{-2} \sin^2 x$ $u = x^{-2} \quad v = \sin^2 x$ $u' = -2x^{-3} \quad v' = 2 \sin x \cos x$ $5e^{3y}(3) \frac{dy}{dx} = x^{-2}(2 \sin x \cos x) + \sin^2 x(-2x^{-3})$ $\frac{dy}{dx} = \frac{2x^{-2} \sin x \cos x - 2x^{-3} \sin^2 x}{5e^{3y}(3)}$ $\frac{dy}{dx} = \frac{2x^{-2} \sin x \cos x - 2x^{-3} \sin^2 x}{x^{-2} \sin^2 x(3)}$ $\frac{dy}{dx} = \frac{2x^{-2} \sin x \cos x}{x^{-2} \sin^2 x(3)} - \frac{2x^{-3} \sin^2 x}{x^{-2} \sin^2 x(3)}$ $\frac{dy}{dx} = \frac{2 \cos x}{3 \sin x} - \frac{2}{3} x^{-1}$ $\frac{dy}{dx} = -\frac{2}{3} \left(\frac{1}{x} \right) + \frac{2}{3} \cot x$	B1 B1 K1 K1 K1 K1J1	u & u' v & v' Differentiate $5e^{3y}$ LHS Product Rule RHS Substitute $5e^{3y} = x^{-2} \sin^2 x$ Simplify and answer

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	$\frac{dy}{dx} = a \left(\frac{1}{x} \right) + b \cot x$ $a = -\frac{2}{3} \text{ and } b = \frac{2}{3}$	J1J1	Value of a and b
Total		9 marks	
3	$f(x) = a \ln x + bx + \frac{6}{x}$ $f'(x) = \frac{a}{x} + b - \frac{6}{x^2}$ $f'(1) = \frac{a}{1} + b - \frac{6}{1}$ $a + b - 6 = 0$ $a + b = 6 \rightarrow (1)$ $f'(-3) = \frac{a}{-3} + b - \frac{6}{(-3)^2}$ $\frac{a}{-3} + b - \frac{6}{9} = 0$ $\frac{a}{-3} + b = \frac{6}{9} \rightarrow (2)$ Solving (1) and (2) $a + b = 6 \rightarrow (1) \quad \frac{a}{-3} + b = \frac{6}{9} \rightarrow (2) \quad a = 4 \text{ and } b = 2$	K1 K1 K1 J1 K1 J1 J1	Differentiate Sub $f'(1)$ and equate to zero Sub $f'(-3)$ and equate to zero Either one correct eqn Solving a and b
Total		7 marks	

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PART B			
1	$z = z = -\sqrt{3} - i$ $ z = \sqrt{(-\sqrt{3})^2 + (-1)^2}$ $R = 4$ $\text{Arg } z = -(\pi - \tan^{-1}(\frac{1}{\sqrt{3}}))$ $\theta = -\frac{5\pi}{6}$ Polar form $Z = 4 \left(\cos\left(-\frac{5\pi}{6}\right) + i \sin\left(-\frac{5\pi}{6}\right) \right)$	K1 K1 J1 K1 J1	Method to find absolute value of z Method to find argument Either absolute value or argument Polar form
Total		5 marks	
2a	$\ln 2^{4x-1} = \ln 8^{x+5} + \log_2 16^{1-2x}$ $(4x-1)\ln 2 = (x+5)\ln 8 + (1-2x)\log_2 16$ $(4x-1)\ln 2 = (x+5)\ln 2^3 + (1-2x)\log_2 2^4$ $(4x-1)\ln 2 = 3(x+5)\ln 2 + 4(1-2x)\log_2 2$ $(4x-1)\ln 2 - 3(x+5)\ln 2 = 4(1-2x)$ $\ln 2[(4x-1) - 3(x+5)] = 4 - 8x$ $\ln 2(x-16) = 4 - 8x$ $x\ln 2 - 16\ln 2 = 4 - 8x$ $x\ln 2 + 8x = 4 + 16\ln 2$ $x = \frac{4 + 16\ln 2}{8 + \ln 2}$	K1 J1 K1 J1	Law of log Linear form Find x Answer in terms of $\ln 2$
2b	$3(x+2) < 7 - 2x$ $3x + 6 < 7 - 2x$	K1	

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	$5x < 1$ $x < \frac{1}{5}$ $\{x: x < \frac{1}{5}\}$	J1	
2b	$\frac{x^2 - 3x - 4}{x + 1} \leq 2$ $\frac{x^2 - 3x - 4}{x + 1} - 2 \leq 0$ $\frac{x^2 - 3x - 4 - 2(x + 1)}{x + 1} \leq 0$ $\frac{x^2 - 5x - 6}{x + 1} \leq 0$ $\frac{(x - 6)(x + 1)}{x + 1} \leq 0$ $x - 6 \leq 0$ $x \leq 6$ Since $x + 1$ is at denominator $x \neq -1$ $\{x: x \leq -1 \text{ or } -1 < x \leq 6\}$	K1 J1	Equate the denominator Simplify Exclude (-1) Answer
	$\{x: x < \frac{1}{5}\}$ and $\{x: x < -1 \text{ or } -1 < x \leq 6\}$ 	K1	Find intersection

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	$\left\{x: -\infty < x < -1 \text{ or } -1 < x < \frac{1}{5}\right\}$	J1	Answer
Total		11 marks	
3a	$f(x) = e^{ax} + b$ and $(g \circ f)(x) = \frac{1}{2} \ln(e^{ax} + c)$ $(f \circ f^{-1})(x) = x$ $e^{af^{-1}(x)} + b = x$ $af^{-1}(x) = \ln(x - b)$ $f^{-1}(x) = \frac{1}{a} \ln(x - b)$ $g \circ (f \circ f^{-1})(x) = \frac{1}{2} \ln\left(e^{a(\frac{1}{a} \ln(x-b))} + c\right)$ $g(x) = \frac{1}{2} \ln(e^{\ln(x-b)} + c)$ $g(x) = \frac{1}{2} \ln(x - b + c) \quad D_{g(x)} = (b - c, \infty)$	K1 K1 J1 K1 K1 J1 J1	Sub $f^{-1}(x)$ into $f(x)$ and equate to x Method to find $f^{-1}(x)$ Answer Sub $f^{-1}(x)$ into $(g \circ f)(x)$ Simplify

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[illegible]

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4	$f(x) = \sqrt{5x+4}$ and $g(x) = \frac{2x-3}{x+1}$ $g(x) = \frac{2x-3}{x+1}; x \neq -1$ $(g \circ g^{-1})(x) = x$ $\frac{2g^{-1}(x)-3}{g^{-1}(x)+1} = x$ $2g^{-1}(x)-3 = xg^{-1}(x)+x$ $g^{-1}(x)[(2-x)] = x+3$ $g^{-1}(x) = \frac{x+3}{2-x}$ $(f \circ g^{-1})(x) = \sqrt{5\left(\frac{x+3}{2-x}\right)+4}$ $(f \circ g^{-1})(0) = \sqrt{5\left(\frac{0+3}{2-0}\right)+4}$ $(f \circ g^{-1})(0) = \frac{\sqrt{46}}{2}$	K1 K1 J1 K1 K1 K1J1	Sub $g^{-1}(x)$ into $g(x)$ and equate to x Method to find $g^{-1}(x)$ Sub $g^{-1}(x)$ into $f(x)$ Sub 0 into $(f \circ g^{-1})(x)$
Total		7 marks	

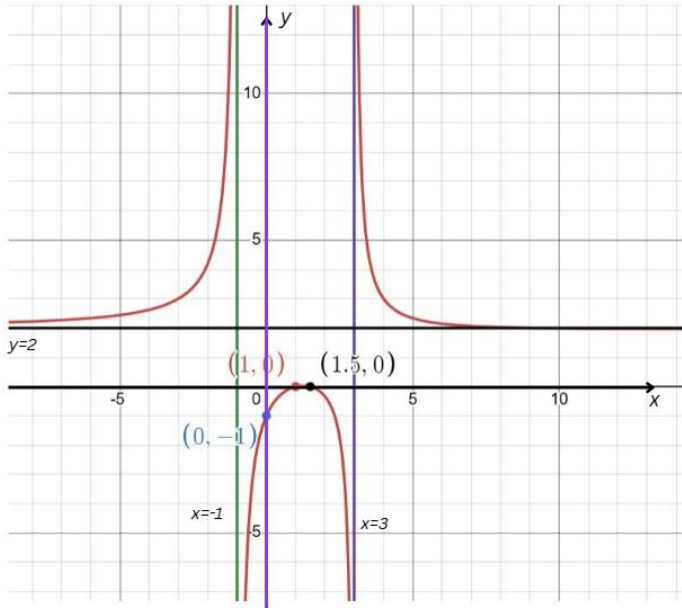
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5a	$f(x) = \frac{2x^2 - 5x + 3}{(x - 3)(x + 1)}$ $x + 1 \neq 0, x - 3 \neq 0$ $x \neq -1, x \neq 3$ $D_{f(x)} = (-\infty, -1) \cup (-1, 3) \cup (3, \infty)$	B1	
5b	$f(x) = \frac{2x^2 - 5x + 3}{(x - 3)(x + 1)} = \frac{(2x - 3)(x - 1)}{(x - 3)(x + 1)}$ x - intercept $y = 0$ $0 = \frac{(2x - 3)(x - 1)}{(x - 3)(x + 1)}$ $x = \frac{3}{2}, \text{ and } x = 1$ x - intercept is $\frac{3}{2}$ and 1 y - intercept $x = 0$ $f(x) = \frac{(2(0) - 3)(0 - 1)}{(0 - 3)(0 + 1)} = -\frac{3}{3} = -1$ y - intercept is -1	K1 J1 K1 J1	
5c	$f(x) = \frac{2x^2 - 5x + 3}{(x - 3)(x + 1)}$ $\lim_{x \rightarrow -1^+} \frac{2x^2 - 5x + 3}{(x - 3)(x + 1)} = \frac{10}{(-3)(0)} = -\infty$	K1	Limit of x approaching -1 from right or left

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	$\lim_{x \rightarrow -1^-} \frac{2x^2 - 5x + 3}{(x - 3)(x + 1)} = \frac{10}{(-3)(0)} = +\infty$ The vertical asymptote is $x = -1$ $\lim_{x \rightarrow 3^+} \frac{2x^2 - 5x + 3}{(x - 3)(x + 1)} = \frac{1}{0} = +\infty$ $\lim_{x \rightarrow 3^-} \frac{2x^2 - 5x + 3}{(x - 3)(x + 1)} = \frac{1}{0} = -\infty$ The vertical asymptote is $x = 3$ $\lim_{x \rightarrow +\infty} \frac{2x^2 - 5x + 3}{x^2 - 2x - 3} = \lim_{x \rightarrow +\infty} \frac{\frac{2x^2 - 5x + 3}{x^2}}{\frac{x^2 - 2x - 3}{x^2}}$ $\lim_{x \rightarrow +\infty} \frac{2 - \frac{5}{x} + \frac{3}{x^2}}{1 - \frac{2}{x} - \frac{3}{x^2}} = \frac{2 - 0 + 0}{1 - 0 - 0} = 2$ $\lim_{x \rightarrow -\infty} \frac{2x^2 - 5x + 3}{x^2 - 2x - 3} = \lim_{x \rightarrow -\infty} \frac{\frac{2x^2 - 5x + 3}{x^2}}{\frac{x^2 - 2x - 3}{x^2}}$ $\lim_{x \rightarrow -\infty} \frac{2 - \frac{5}{x} + \frac{3}{x^2}}{1 - \frac{2}{x} - \frac{3}{x^2}} = \frac{2 - 0 + 0}{1 - 0 - 0} = 2$ The horizontal asymptote is $y = 2$	J1 K1 J1 K1 J1	Limit of x approaching 2 from right or left Limit of x approaching $+\infty$ or $-\infty$

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NO.	SOLUTION		MARKS	REMARKS
5d			R1 R1 R1	Shape Label intersection points and asymptotes All correct including label x and y axis
Total			14 marks	
6a	$y = x + \ln(\cos y)$ $\frac{dy}{dx} = 1 + \frac{1}{\cos y} (-\sin y) \frac{dy}{dx}$ $\frac{dy}{dx} + \left(\frac{\sin y}{\cos y}\right) \frac{dy}{dx} = 1$ $\frac{dy}{dx} (1 + \tan y) = 1$ $\frac{dy}{dx} = \frac{1}{1 + \tan y}$	$\frac{dy}{dx} = 1 + \frac{1}{\cos y} (-\sin y) \frac{dy}{dx}$ $\frac{dy}{dx} + \left(\frac{\sin y}{\cos y}\right) \frac{dy}{dx} = 1$ $\cos y \frac{dy}{dx} + \sin y \frac{dy}{dx} = \cos y$ $\frac{dy}{dx} = \frac{\cos y}{\cos y + \sin y}$	K1 K1 K1 J1	Differentiate $\ln(\cos y)$ and $\cos y$ Correct u, v and Correct u', v' Try to find dy/dx

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	$\frac{dy}{dx} = (1 + \tan y)^{-1}$ $\frac{d^2y}{dx^2} = -(1 + \tan y)^{-2} \sec^2 y \frac{dy}{dx}$ $\frac{d^2y}{dx^2} = -\frac{\sec^2 y}{(1 + \tan y)^2} \frac{dy}{dx}$ $\frac{d^2y}{dx^2} = -\frac{\sec^2 y}{(1 + \tan y)^3}$ $\frac{d^2y}{dx^2} + \frac{\sec^2 y}{(1 + \tan y)^3} = 0$	K1 K1 K1 J1	Differentiate using power rule Differentiate trigo Sub $\frac{dy}{dx}$ Try to prove
6b	$x = 2t^2 - 3t \quad y = \frac{t^2 + 2t}{t - 1}$ $\frac{dx}{dt} = 4t - 3 \quad u = t^2 + 2t \quad v = t - 1$ $u' = 2t + 2 \quad v' = 1$ $\frac{dy}{dt} = \frac{(t-1)(2t+2) - (t^2+2t)}{(t-1)^2} = \frac{t^2 - 2t - 2}{(t-1)^2}$ $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{t^2 - 2t - 2}{(t-1)^2} \times \frac{1}{4t - 3} = \frac{t^2 - 2t - 2}{(t-1)^2(4t - 3)}$ When $t = 2$	K1 K1 K1 J1	Correct $\frac{dx}{dt}$ Correct $\frac{dy}{dt}$ Method to find $\frac{dy}{dx}$ Correct $\frac{dy}{dx}$

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	$x = 2(2)^2 - 3(2) = 2$ $y = \frac{t^2 + 2t}{t - 1} = \frac{2^2 + 2(2)}{2 - 1} = 8$ $\frac{dy}{dx} = \frac{t^2 - 2t - 2}{(t - 1)^2(4t - 3)} = \frac{2^2 - 2(2) - 2}{(2 - 1)^2(4(2) - 3)} = -\frac{2}{5}$ $y - 8 = -\frac{2}{5}(x - 2)$ $y = -\frac{2}{5}x + \frac{44}{5}$	K1 J1 K1 J1	Find x and y Value of x and y Find $\frac{dy}{dx}$ Tangent equation
Total		16 Marks	
7	$2x + 3y = 60$ $y = \frac{60 - 2x}{3}$ $P = x^2 \left(\frac{60 - 2x}{3} \right)$ $P = \frac{2}{3}(30x^2 - x^3)$ $\frac{dP}{dx} = \frac{2}{3}(60x - 3x^2)$ $\frac{2}{3}(60x - 3x^2) = 0$ $3x(20 - x) = 0$	K1 K1 K1	P in terms of x or y $\frac{dP}{dx}$ $\frac{dP}{dx} = 0$

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	$x = 0, x = 20$ $\frac{d^2P}{dx^2} = \frac{2}{3}(60 - 6x)$ <i>when $x = 0$</i> <i>when $x = 20$</i> $\frac{d^2P}{dx^2} = 40 > 0$ P is min $\frac{d^2P}{dx^2} = -40 < 0$ P is max P is maximum when $x = 20$ and $y = \frac{60 - 2x}{3} = \frac{60 - 2(20)}{3} = \frac{20}{3}$ $P = x^2y = (20^2)\left(\frac{20}{3}\right) = \frac{8000}{3}$	J1 K1 J1 J1 K1 J1	Value of x or y $\frac{d^2P}{dx^2}$ Testing value of x or y Value of y Sub x and y into P , value of P
Total		9 marks	
7	$2x + 3y = 60$ $x = \frac{60 - 3y}{2}$ and $P = \left(\frac{60 - 3y}{2}\right)^2 (y)$ $P = \frac{9}{4}(400y - 40y^2 + y^3)$ $\frac{dP}{dy} = \frac{9}{4}(400 - 80y + 3y^2)$ $\frac{9}{4}(400 - 80y + 3y^2) = 0$ $y = 20, y = \frac{20}{3}$	 K1 K1 K1	 P in terms of x or y $\frac{dP}{dx}$ or $\frac{dP}{dy}$ $\frac{dP}{dx} = 0$ or $\frac{dP}{dy} = 0$

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	$\frac{d^2P}{dy^2} = \frac{9}{4}(6y - 80)$ $\text{when } y = 20$ $\frac{d^2P}{dy^2} = 90 > 0 \text{ } P \text{ is min}$ $P \text{ is maximum when } y = \frac{20}{3} \text{ and}$ $x = \frac{60 - 3y}{2} = \frac{60 - 3(\frac{20}{3})}{2} = 20$ $P = x^2y = (20^2)\left(\frac{20}{3}\right) = \frac{8000}{3}$	J1 K1 ketat √K1 J1 √K1 J1	Value of x or y $\frac{d^2P}{dx^2}$ Testing his value of x or y (have 2 values to test) Value of y when x=20 Sub his x and y into P, value of P
Total		9 marks	